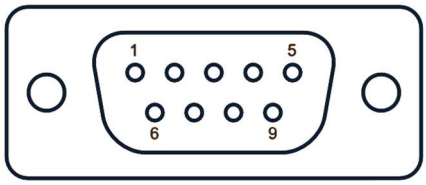
Electrical Specifications

Electrical Interface Connector

Table 1: BEE-DAL Electrical Interface Connector Information

Item	Description
Connector type	Sub D 9 pin vacuum connector
Connector Model / Manufacturer	FA13603 / Solid Sealing Technology
Mating Feedthrough / Manufacturer	FA13118 / Solid Sealing Technology



Current rating: max. 5A per contact

Table 2: BEE-DAL Interface Connector Pin Description

Pin	Name	Description
1	IO	BEE-DAL IO debug pin
2	RX- / B	BEE-DAL RS-422 RX- inverting receiver input
3	TX+ / Y	BEE-DAL RS-422 TX+ driver output
4	GND	GND
5	GND	GND
6	RX+ / A	BEE-DAL RS-422 RX+ receiver input
7	TX- / Z	BEE-DAL RS-422 TX- inverting driver output
8	VIN	Input voltage
9	VIN	Input voltage

Power Interface

Table 3: BEE-DAL Electrical Power Characteristics

Parameter	Min.	Typ.	Max.	Unit
Supply voltage	11.2	12.0	13.5	V
Current consumption	300	1000	1250	mA
Power consumption	3.6	12.0	[1]15.0	W
[2]Inrush current	-	-	1500	mA

[1]Payload interlock limits maximum power consumption to 60 seconds

[2]Inrush current maximum duration is 5 seconds

Table 4: BEE-DAL Power Consumption Analysis

Payload Mode	Average Power	Peak Power	Duration / Duty Cycle
Off / safe state	0	-	-
Boot / initialization	3 W	8 W	60s / 100%
Standby	5 W	-	- / 100%
Nominal operation	12 W	15 W	- / 100%
Data transfer	8 W	10 W	- / 100%
Fault / recovery	TBD	TBD	TBD
Ground checkout	TBD	TBD	TBD

Signal Interface

Table 5: BEE-DAL Signal Interface Characteristics

Parameter		Min.	Typ.	Max.	Unit
RS-422	Transceiver	Analog Devices LTC2865			
	Operational voltage	3.0	-	5.5	V
	ESD protection	±15kV			-
	Datarate	-	921,600	16,000,000	bps
	Protocol	Native, KISS			-

TMTC and Data Transfer

RS-422 configuration

BEE-DAL payload TMTC communication, which operates over RS-422 with below configurations.

UART configuration

- 8N1, parity none, baud rate **921600**
- **LSB** transmitted first

Transport protocol configuration

- Multiple bytes order: **Little Endian**
- Package timeout: **2000 ms**

Package format

TC package format

Name	START	TYPE	ADDRESS	LENGTH	PAYLOAD	CRC-16
Size (B)	1	1	1	2	N	2
Byte offset	0	1	2	3	5	5+N

START byte: **0xA5**

TYPE:

- **0x00**: PING
- **0x01**: System configuration
- **0x02**: Experiment control
- **0x03**: Data transfer
- **0x04**: Payload telemetry
- **0x05**: SFU

LENGTH: 0-65535

- If LENGTH = 0, CRC-16 shall be **0xFFFF**

TM package format

Name	START	ACK / ERR	ADDRESS / ERRCODE	LENGTH	PAYLOAD	CRC-16
Size (B)	1	1	1	2	N	2
Byte offset	0	1	2	3	5	5+N

ACK / ERR:

- ACK: copy of TC TYPE and ADDRESS
- ERR: **0xFF**

ERRCODE:

- **0x00**: TC invalid
- **0x01**: Data error
- **0x02**: Memory error
- **0x03**: Unknown error

TM ACK package

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	TC TYPE
ADDR/ERRCODE	U8	1	2	TC ADDR
LENGTH	U16	2	3	4
System uptime (sec)	U32	4	5	-
CRC-16	U16	2	9	-

CRC-16

CRC-16 CCSDS

- Polynomial: **0x1021**
- Initial value: **0xFFFF**
- In/out reflection: **false/false**
- Output XOR value: **0x0000**

CRC-32

CRC-32 ISO-HDLC

- Polynomial: **0x04C11DB7**
- Initial value: **0xFFFFFFFF**
- In/out reflection: **true/true**
- Output XOR value: **0xFFFFFFFF**

TMTC Summary

Table 8: BEE-DAL TMTC Summary

Type	Address	Name	Description	Response
0x00	0x00	ping	Communication check	ACK
0x01	0x00	sys_info	System information	info data
0x01	0x01	sys_rtc_cfg	Configure system clock	current epoch
0x01	0x02	pwr_adj_cfg	Configure adjustable power	power config status
0x01	0x03	pwr_en	Enable/Disable power lines	power en status
0x01	0x04	pwr_stat	Get system power statuses	power status
0x01	0x05	sen_bme_cfg	Sensor BME configuration	BME config
0x01	0x06	sen_imu_cfg	Sensor IMU configuration	IMU config
0x01	0x07	sen_stat	Sensor status	Sensor status
0x01	0x10	uvc_cfg	UVC configuration	UVC config
0x01	0x11	uvc_ctl	UVC control	ACK
0x01	0x12	uvc_stat	UVC status	UVC status
0x01	0x13	thm_cfg	Thermal configuration	Thermal config
0x01	0x14	thm_ctl	Thermal control	ACK
0x01	0x15	thm_stat	Thermal status	Thermal status
0x01	0x16	fld_cfg	Fluidic configuration	Fluidic config
0x01	0x17	fld_ctl	Fluidic control	ACK
0x01	0x18	fld_stat	Fluidic status	Fluidic status
0x01	0x19	rba_cfg	RBA configuration	RBA config
0x01	0x1A	rba_ctl	RBA control	ACK
0x01	0x1B	rba_stat	RBA status	RBA status
0x01	0x20	lev_calib	LEV calibration	LEV calib data
0x01	0x21	lev_cfg	LEV configuration	LEV config
0x01	0x22	lev_ctl	LEV control	ACK
0x01	0x23	lev_stat	LEV status	LEV status
0x02	0x00	exp_stat	EXPeriment status	EXP status
0x02	0x01	exp_cfg	EXP configuration	EXP config
0x02	0x02	exp_ctl	EXP control	ACK
0x02	0x03	thp_cfg	Thermal profile configuration	THP config
0x02	0x04	thp_en	Thermal profile enable	ACK
0x02	0x05	mon_cfg	Monitoring configuration	MON config
0x02	0x06	mon_en	Monitoring enable	ACK
0x02	0x07	rsv_op1	Reserved operation #1	ACK
0x02	0x08	rsv_op2	Reserved operation #2	RSV #2 response
0x02	0x09	rsv_op3	Reserved operation #3	RSV #3 response
0x03	0x00	dat_lst	Get number of data files	count of data files by type
0x03	0x01	dat_sel	Select file to download	file metadata
0x03	0x02	dat_get	Download (chunks) data	(chunks) data
0x03	0x03	dat_des	Release current file	ACK
0x03	0x04	dat_del	Delete file data	count of data files

0x03	0x05	dat_mrk	Mark data file	ACK
0x04	0x00	tm_get	Payload telemetry get	TM data
0x04	0x01	tm_cfg	Payload TM configuration	TM config

NOTE: This summary table supposes to be updated.

TMTC Tables

PING (TYPE = 0x00)

Communication check.

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x00
ADDR	U8	1	2	0x00
LENGTH	U16	2	3	0
CRC-16	U16	2	5	0xFFFF

TM

TM ACK package

System Configuration (TYPE = 0x01)

sys_info (0x00)

Get system information.

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x01
ADDR	U8	1	2	0x00
LENGTH	U16	2	3	[1]0/1
TC Payload	U8	0/1	5	-
CRC-16	U16	2	5/6	-

[1]TC LENGTH:

- 0: get command
- 1: set command, write 0x1 to TC Payload will perform a system self-test

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x01
ADDR/ERRCODE	U8	1	2	0x00
LENGTH	U16	2	3	96
TM Payload	U8	96	5	-
CRC-16	U16	2	101	-

sys_rtc_cfg (0x01)

Configure system clock

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x01
ADDR	U8	1	2	0x01
LENGTH	U16	2	3	[1]0/4
Unix epoch (UTC)	U32	0/4	5	-
CRC-16	U16	2	9	-

[1]TC LENGTH:

- 0: get command
- 4: set command with Unix epoch (4B)

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x01
ADDR/ERRCODE	U8	1	2	0x01
LENGTH	U16	2	3	4
System epoch	U32	4	5	-
CRC-16	U16	2	9	-

pwr_adj_cfg (0x02)

Configure adjustable power

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x01
ADDR	U8	1	2	0x02
LENGTH	U16	2	3	[1]0/8
TC Payload	U8	0/8	5	-
CRC-16	U16	2	5/13	-

[1]TC LENGTH:

- 0: get command
- 8: set command with 8B TC payload

TM

	Data Type	Size (B)	Byte offset	Value

START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x01
ADDR/ERRCODE	U8	1	2	0x02
LENGTH	U16	2	3	8
TM Payload	U8	8	5	-
CRC-16	U16	2	13	-

pwr_en (0x03)

Enable/Disable power lines

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x01
ADDR	U8	1	2	0x03
LENGTH	U16	2	3	[1]0/2
TC Payload	U16	0/2	5	-
CRC-16	U16	2	5/7	-

[1]TC LENGTH:

- 0: get command
- 2: set command with 2B TC payload

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x01
ADDR/ERRCODE	U8	1	2	0x03
LENGTH	U16	2	3	2
TM Payload	U16	2	5	-
CRC-16	U16	2	7	-

pwr_stat (0x04)

Get system power status

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x01
ADDR	U8	1	2	0x04
LENGTH	U16	2	3	2
[1]Power report type	U8	1	5	-

CRC-16	U16	2	6	-

[1]Power report type:

- 0: All statuses
- 1: Monitor status
- 2: Enable status
- 3: Error status

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x01
ADDR/ERRCODE	U8	1	2	0x04
LENGTH	U16	2	3	N
TM Payload	U16	N	5	-
CRC-16	U16	2	5+N	-

sen_bme_cfg (0x05)

Configure BME sensor

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x01
ADDR	U8	1	2	0x05
LENGTH	U16	2	3	[1]0/8
TC Payload	U8	0/8	5	-
CRC-16	U16	2	5/13	-

[1]TC LENGTH:

- 0: get command
- 8: set command with 8B TC payload

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x01
ADDR/ERRCODE	U8	1	2	0x05
LENGTH	U16	2	3	8
TM Payload	U8	8	5	-
CRC-16	U16	2	13	-

sen_imu_cfg (0x06)

Configure IMU sensor

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x01
ADDR	U8	1	2	0x06
LENGTH	U16	2	3	[1]0/8
TC Payload	U8	0/8	5	-
CRC-16	U16	2	5/13	-

[1]TC LENGTH:

- 0: get command
- 8: set command with 8B TC payload

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x01
ADDR/ERRCODE	U8	1	2	0x06
LENGTH	U16	2	3	8
TM Payload	U8	8	5	-
CRC-16	U16	2	13	-

sen_stat (0x07)

Get system power status

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x01
ADDR	U8	1	2	0x07
LENGTH	U16	2	3	2
[1]Sensor report type	U8	1	5	-
CRC-16	U16	2	6	-

[1]Sensor report Type:

- 0: All sensor data
- 1: Temperature sensor data
- 2: BME sensor data
- 3: IMU sensor data
- 4: Flow sensor data

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x01
ADDR/ERRCODE	U8	1	2	0x07
LENGTH	U16	2	3	N
TM Payload	I16	N	5	-
CRC-16	U16	2	5+N	-

uvc_cfg (0x10)

Configure UVC subsystem

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x01
ADDR	U8	1	2	0x10
LENGTH	U16	2	3	[1]0/32
TC Payload	U8	0/32	5	-
CRC-16	U16	2	5/37	-

[1]TC LENGTH:

- 0: get command
- 32: set command with 32B TC payload

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x01
ADDR/ERRCODE	U8	1	2	0x10
LENGTH	U16	2	3	32
TM Payload	U8	32	5	-
CRC-16	U16	2	37	-

uvc_ctl (0x11)

UVC subsystem manual control

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x01
ADDR	U8	1	2	0x11

LENGTH	U16	2	3	8
TC Payload	U8	8	5	-
CRC-16	U16	2	13	-

TM

TM ACK package

uvc_stat (0x12)

Get UVC subsystem status

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x01
ADDR	U8	1	2	0x12
LENGTH	U16	2	3	0
CRC-16	U16	2	5	-

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x01
ADDR/ERRCODE	U8	1	2	0x12
LENGTH	U16	2	3	6
UVC status	U16	6	5	-
CRC-16	U16	2	11	-

thm_cfg (0x13)

Configure thermal subsystem

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x01
ADDR	U8	1	2	0x13
LENGTH	U16	2	3	[1]0/32
TC Payload	U8	0/32	5	-
CRC-16	U16	2	5/37	-

[1]TC LENGTH:

- 0: get command

- 32: set command with 32B TC payload

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x01
ADDR/ERRCODE	U8	1	2	0x13
LENGTH	U16	2	3	32
TM Payload	U8	32	5	-
CRC-16	U16	2	37	-

thm_ctl (0x14)

Thermal subsystem manual control

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x01
ADDR	U8	1	2	0x14
LENGTH	U16	2	3	8
TC Payload	U8	8	5	-
CRC-16	U16	2	13	-

TM

TM ACK package

thm_stat (0x15)

Get thermal subsystem status

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x01
ADDR	U8	1	2	0x15
LENGTH	U16	2	3	0
CRC-16	U16	2	5	-

TM

	Data Type	Size (B)	Byte offset	Value

START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x01
ADDR/ERRCODE	U8	1	2	0x15
LENGTH	U16	2	3	6
THM status	U16	6	5	-
CRC-16	U16	2	11	-

fld_cfg (0x16)

Configure fluidic subsystem

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x01
ADDR	U8	1	2	0x16
LENGTH	U16	2	3	[1]0/32
TC Payload	U8	0/32	5	-
CRC-16	U16	2	5/37	-

[1]TC LENGTH:

- 0: get command
- 32: set command with 32B TC payload

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x01
ADDR/ERRCODE	U8	1	2	0x16
LENGTH	U16	2	3	32
TM Payload	U8	32	5	-
CRC-16	U16	2	37	-

fld_ctl (0x17)

Fluidic subsystem manual control

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x01
ADDR	U8	1	2	0x17
LENGTH	U16	2	3	8
TC Payload	U8	8	5	-
CRC-16	U16	2	13	-

TM

TM ACK package

fld_stat (0x18)

Get fluidic subsystem status

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x01
ADDR	U8	1	2	0x18
LENGTH	U16	2	3	0
CRC-16	U16	2	5	-

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x01
ADDR/ERRCODE	U8	1	2	0x18
LENGTH	U16	2	3	16
FLD status	U16	16	5	-
CRC-16	U16	2	21	-

rba_cfg (0x19)

Configure RBA subsystem

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x01
ADDR	U8	1	2	0x19
LENGTH	U16	2	3	[1]0/32
TC Payload	U8	0/32	5	-
CRC-16	U16	2	5/37	-

[1]TC LENGTH:

- 0: get command
- 32: set command with 32B TC payload

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x01
ADDR/ERRCODE	U8	1	2	0x19
LENGTH	U16	2	3	32
TM Payload	U8	32	5	-
CRC-16	U16	2	37	-

rba_ctl (0x1A)

RBA subsystem manual control

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x01
ADDR	U8	1	2	0x1A
LENGTH	U16	2	3	8
TC Payload	U8	8	5	-
CRC-16	U16	2	13	-

TM

TM ACK package

rba_stat (0x1B)

Get fluidic subsystem status

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x01
ADDR	U8	1	2	0x1B
LENGTH	U16	2	3	0
CRC-16	U16	2	5	-

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x01
ADDR/ERRCODE	U8	1	2	0x1B
LENGTH	U16	2	3	16

RBA status	U16	16	5	-
CRC-16	U16	2	21	-

lev_calib (0x20)

Calib LEV subsystem

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x01
ADDR	U8	1	2	0x21
LENGTH	U16	2	3	[1]0/130
TC Payload	U8	0/130	5	-
CRC-16	U16	2	5/135	-

[1]TC LENGTH:

- 0: get command
- 128: set command with 128B TC payload

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x01
ADDR/ERRCODE	U8	1	2	0x21
LENGTH	U16	2	3	130
TM Payload	U8	130	5	-
CRC-16	U16	2	135	-

lev_cfg (0x21)

Configure LEV subsystem

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x01
ADDR	U8	1	2	0x21
LENGTH	U16	2	3	[1]0/32
TC Payload	U8	0/32	5	-
CRC-16	U16	2	5/37	-

[1]TC LENGTH:

- 0: get command
- 32: set command with 32B TC payload

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x01
ADDR/ERRCODE	U8	1	2	0x21
LENGTH	U16	2	3	32
TM Payload	U8	32	5	-
CRC-16	U16	2	37	-

lev_ctl (0x22)

LEV subsystem manual control

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x01
ADDR	U8	1	2	0x22
LENGTH	U16	2	3	8
TC Payload	U8	8	5	-
CRC-16	U16	2	13	-

TM

TM ACK package

lev_stat (0x23)

Get LEV subsystem status

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x01
ADDR	U8	1	2	0x23
LENGTH	U16	2	3	0
CRC-16	U16	2	5	-

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x01

ADDR/ERRCODE	U8	1	2	0x23
LENGTH	U16	2	3	16
LEV status	U16	16	5	-
CRC-16	U16	2	21	-

Experiment Control (TYPE = 0x02)

exp_stat (0x00)

Get EXPeriment status

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x02
ADDR	U8	1	2	0x00
LENGTH	U16	2	3	0
CRC-16	U16	2	5	-

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x02
ADDR/ERRCODE	U8	1	2	0x00
LENGTH	U16	2	3	1
EXP status	U8	1	5	-
CRC-16	U16	2	6	-

exp_cfg (0x01)

EXP configure

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x02
ADDR	U8	1	2	0x01
LENGTH	U16	2	3	[1]0/32
TC Payload	U8	0/32	5	-
CRC-16	U16	2	5/37	-

[1]TC LENGTH:

- 0: get command

- 32: set command with 32B TC payload

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x02
ADDR/ERRCODE	U8	1	2	0x01
LENGTH	U16	2	3	32
TM Payload	U8	32	5	-
CRC-16	U16	2	37	-

exp_ctl (0x02)

EXP control

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x02
ADDR	U8	1	2	0x02
LENGTH	U16	2	3	8
TC Payload	U8	8	5	-
CRC-16	U16	2	13	-

TM

TM ACK package

thp_cfg (0x03)

Thermal profile configure

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x02
ADDR	U8	1	2	0x03
LENGTH	U16	2	3	[1]0/32
TC Payload	U8	0/32	5	-
CRC-16	U16	2	5/37	-

[1]TC LENGTH:

- 0: get command
- 32: set command with 32B TC payload

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x02
ADDR/ERRCODE	U8	1	2	0x03
LENGTH	U16	2	3	32
TM Payload	U8	32	5	-
CRC-16	U16	2	37	-

thp_en (0x04)

Thermal profile enable

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x02
ADDR	U8	1	2	0x04
LENGTH	U16	2	3	[1]0/1
TC Payload	U8	0/1	5	-
CRC-16	U16	2	6	-

[1]TC LENGTH:

- 0: get command
- 1: set command with 1B TC payload

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x02
ADDR/ERRCODE	U8	1	2	0x04
LENGTH	U16	2	3	1
TM Payload	U8	1	5	-
CRC-16	U16	2	6	-

mon_cfg (0x05)

Monitoring configure

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x02

ADDR	U8	1	2	0x05
LENGTH	U16	2	3	[1]0/32
TC Payload	U8	0/32	5	-
CRC-16	U16	2	5/37	-

[1]TC LENGTH:

- 0: get command
- 32: set command with 32B TC payload

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x02
ADDR/ERRCODE	U8	1	2	0x05
LENGTH	U16	2	3	32
TM Payload	U8	32	5	-
CRC-16	U16	2	37	-

mon_en (0x06)

Monitoring enable

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x02
ADDR	U8	1	2	0x06
LENGTH	U16	2	3	[1]0/1
TC Payload	U8	0/1	5	-
CRC-16	U16	2	6	-

[1]TC LENGTH:

- 0: get command
- 1: set command with 1B TC payload

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x02
ADDR/ERRCODE	U8	1	2	0x06
LENGTH	U16	2	3	1
TM Payload	U8	1	5	-
CRC-16	U16	2	6	-

rsv_op1 (0x07)

LEV subsystem manual control

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x02
ADDR	U8	1	2	0x07
LENGTH	U16	2	3	32
TC Payload	U8	32	5	-
CRC-16	U16	2	37	-

TM

TM ACK package

rsv_op2 (0x08)

Monitoring configure

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x02
ADDR	U8	1	2	0x08
LENGTH	U16	2	3	32
TC Payload	U8	32	5	-
CRC-16	U16	2	37	-

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x02
ADDR/ERRCODE	U8	1	2	0x08
LENGTH	U16	2	3	32
TM Payload	U8	32	5	-
CRC-16	U16	2	37	-

rsv_op3 (0x09)

Monitoring configure

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x02
ADDR	U8	1	2	0x09
LENGTH	U16	2	3	[1]0/32
TC Payload	U8	0/32	5	-
CRC-16	U16	2	5/37	-

[1]TC LENGTH:

- 0: get command
- 32: set command with 32B TC payload

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x02
ADDR/ERRCODE	U8	1	2	0x09
LENGTH	U16	2	3	32
TM Payload	U8	32	5	-
CRC-16	U16	2	37	-

Data Transfer (TYPE = 0x03)

dat_lst (0x00)

List data files by type, with/without mark filter.

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x03
ADDR	U8	1	2	0x00
LENGTH	U16	2	3	1
[1]Mark filter	U8	1	5	0xFF
CRC-16	U16	2	6	-

[1]Mark filter:

- 0xFF: no filter (list all files)
- 0x00: unmarked files only
- 0x01-0xFE: mark value

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x03

ADDR/ERRCODE	U8	1	2	0x00
LENGTH	U16	2	3	12
HK file count	U32	4	5	-
SCI1 file count	U32	4	9	-
SCI2 file count	U32	4	13	-
CRC-16	U16	2	17	-

dat_sel (0x01)

Select data file to process.

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x03
ADDR	U8	1	2	0x01
LENGTH	U16	2	3	5
[1]Data Type	U8	1	5	0-2
Data Index	U16	2	6	0 – 65535
Chunk size (bytes)	U16	2	8	1 – 65535
CRC-16	U16	2	10	-

[1]Data Type:

- 0: HK
- 1: SCI1
- 2: SCI2

Chunk size is limit to 32768B.

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x03
ADDR/ERRCODE	U8	1	2	0x01
LENGTH	U16	2	3	41
Data Type	U8	1	5	-
Data Index	U32	4	6	-
Chunk size (bytes)	U16	4	10	-
File name	STR(20)	20	14	-
File size (bytes)	U32	4	34	-
[1]File CRC-32	U32	4	38	-
[2]Chunk count	U32	4	42	-
CRC-16	U16	2	46	-

[1]File CRC-32: CRC-32 ISO-HDLC

[2]Chunk count = ROUNDUP(File size / Chunk size, 0)

dat_get (0x02)

Transfer data chunk of selected data file.

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x03
ADDR	U8	1	2	0x02
LENGTH	U16	2	3	4
Chunk Index	U32	4	5	$0 – (2^{32}-1)$
CRC-16	U16	2	9	-

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x03
ADDR/ERRCODE	U8	1	2	0x02
LENGTH	U16	2	3	N
Data Chunk	U8	N	5	-
CRC-16	U16	2	5 + N	-

dat_des (0x03)

Deselect current selected data file.

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x03
ADDR	U8	1	2	0x03
LENGTH	U16	2	3	0
CRC-16	U16	2	5	0xFFFF

TM

TM ACK package

dat_del (0x04)

Delele data file at index, by type.

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x03
ADDR	U8	1	2	0x04
LENGTH	U16	2	3	5
Data Type	U8	1	5	-
Data Index	U32	4	6	-
CRC-16	U16	2	10	-

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x03
ADDR/ERRCODE	U8	1	2	0x04
LENGTH	U16	2	3	12
HK Count	U32	4	5	-
SCI1 Count	U32	4	9	-
SCI2 Count	U32	4	13	-
CRC-16	U16	2	17	-

dat_mrk (0x05)

Mark/label current selected data file.

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x03
ADDR	U8	1	2	0x05
LENGTH	U16	2	3	1
Mark Value	U8	1	5	-
CRC-16	U16	2	6	-

TM

TM ACK package

Telemetry (TYPE = 0x04)

tm_get (0x00)

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x04
ADDR	U8	1	2	0x00
LENGTH	U16	2	3	0
CRC-16	U16	2	5	0xFFFF

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x04
ADDR/ERRCODE	U8	1	2	0x00
LENGTH	U16	2	3	29
System time (Linux epoch)	U32	4	5	-
System status	I8	1	9	-
Bootcount	U32	4	10	-
Onboard Temperature	I16	6	14	[pdu, exp, obc]
NTC Temperature	I16	8	20	NTC[0,1,2,3]
MLX Avg Temp	I16	2	28	
BME Temperature	I16	2	30	-
BME Pressure	U16	2	32	-
BME Humidity	U16	2	34	-
IMU Sensor Data	I16	14	36	[t,ax,ay,az,gx,gy,gz]
SLF1 Sensor Data	I16	4	50	[temp, flow]
SLF2 Sensor Data	I16	4	54	[temp, flow]
RBA Status	U16	2	58	-
TA Status	U16	2	60	-
CRC-16	U16	2	62	-

tm_cfg (0x01)

Telemetry configure

TC

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
TYPE	U8	1	1	0x04
ADDR	U8	1	2	0x01
LENGTH	U16	2	3	[1]0/8
TC Payload	U8	0/8	5	-
CRC-16	U16	2	5/13	-

[1]TC LENGTH:

- 0: get command
- 8: set command with 8B TC payload

TM

	Data Type	Size (B)	Byte offset	Value
START	U8	1	0	0xA5
ACK/ERR	U8	1	1	0x04
ADDR/ERRCODE	U8	1	2	0x01
LENGTH	U16	2	3	8
TM Payload	U8	8	5	-
CRC-16	U16	2	13	-

Data transfer

Data Description

Data Type	Size	Format
HK	10 KB per file	Binary
SCI1	1-2 MB per file	Image, .JPG
SCI2	100 KB per file	Binary

Example of payload data transfer as follows.

